

SAFFRON: A Physics-Informed, Label-Free Self-Supervised Deep Learning Framework for Fast and Accurate 3D Fetal Brain MRI Reconstruction

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Abstract—Fetal brain MRI has become indispensable in prenatal diagnosis, offering unique soft tissue contrast to evaluate cortical development and detect neurological abnormalities. While high-resolution 3D imaging can provide valuable anatomical information, clinical acquisitions are often inadequate for reliable volumetric reconstruction due to unpredictable fetal motion and thick-slice protocols. Conventional iterative reconstruction methods, though effective, are computationally intensive and struggle with severe motion artifacts, limiting their feasibility in fast-paced clinical workflows. At the same time, the scarcity of authentic 3D fetal brain volumes prevents supervised learning approaches from developing generalizable reconstruction models. To overcome these limitations, we introduce SAFFRON, a physics-informed, label-free self-supervised framework for efficient and high-fidelity 3D fetal brain MRI reconstruction. SAFFRON eliminates the need for ground-truth 3D volumes by combining physics-driven slice acquisition modeling with data-driven deep learning, thereby bridging model-based and learning-based paradigms. The reconstruction task is decomposed into two modules: (1) multi-stack motion estimation via an SVR network that aligns slices into a canonical space, and (2) 3D volume reconstruction via an SRR network. Reconstruction quality is further enhanced by two targeted constraints: a stack-level contextual consistency loss to guide more accurate alignment and a slice-level adversarial loss to promote anatomically realistic structures. Extensive experiments on simulated and clinical datasets demonstrate that SAFFRON substantially outperforms state-of-the-art methods, achieving superior reconstruction accuracy while delivering up to a $60\times$ acceleration in processing speed.

Index Terms—Fetal brain MRI, slice-to-volume image reconstruction, motion correction, self-supervised learning.

I. INTRODUCTION

A. Motivation

FETAL brain MRI has become indispensable in prenatal diagnosis, providing unparalleled soft tissue contrast to evaluate cortical development and identify neurological abnormalities [1], [2], [3]. While high-resolution 3D imaging can provide more detailed and accurate anatomical information, which may assist in diagnosing fetal brain malformations and assessing developmental trajectories, clinical scan data are often inadequate for reliable volumetric reconstruction due to unpredictable fetal motion and limited resolution. Standard clinical protocols typically acquire multiple thick slice stacks in varying orientations for volumetric reconstruction, using sequences such as single shot fast spin echo (SSFSE) [1]. However, misalignments between slices and low through-plane resolution can degrade volume reconstruction quality. In clinical settings, where high-throughput imaging is essential, the slice thicknesses are typically around 3 mm or slightly larger to improve the signal-to-noise ratio (SNR) and reduce scan time [4], further complicating the precision of the reconstruction. Reliable and efficient reconstruction methods are therefore critical to meet the demands of fast-paced clinical workflows, enabling both accurate pathology analysis and developmental studies.

Traditional reconstruction techniques rely on iterative slice-to-volume registration (SVR) for motion correction, followed by super-resolution reconstruction (SRR) [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. While these methods are effective, they are computationally intensive and struggle with severe motion artifacts in thick-slice images. To address these limitations, data-driven methods have emerged for fetal brain MRI reconstruction [15], [16], [17], [18], [19], [20], [21]. While deep learning enables fast 3D reconstruction, its clinical translation for fetal MRI is hampered by model training on synthetic data [22], a necessary but problematic substitute given the extreme scarcity of real 3D scan fetal brain datasets. Self-supervised approaches [23], [24] provide an alternative solution

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by reconstructing 3D volumes directly from 2D slices through physics-based slice acquisition models. Although demonstrating promising results, these methods still face two key limitations: (1) dependence on accurate initial slice alignment, and (2) high computational costs, ranging from several hours in convolutional neural network (CNN)-based subject-specific training frameworks [24] to several minutes in recent optimized MLP-based subject-specific training frameworks [23], which still hinder seamless clinical deployment.

To overcome these challenges, we propose *SAFFRON*, a physics-informed, label-free self-supervised deep Learning framework for Fast and accurate 3D Fetal brain MRI Reconstruction. SAFFRON reformulates the reconstruction task into two decoupled and efficient modules: (1) multi-stack motion estimation and (2) 3D volume reconstruction. For motion estimation, an SVR network learns rigid transformations that align 2D slices from multiple stacks into a canonical space. For volume reconstruction, an SRR network generates the 3D volume from the aligned slices. To further enhance performance, we introduce two targeted strategies. First, a stack-level contextual consistency loss exploits cross-view similarity to guide more accurate slice alignment. Second, a slice-level adversarial constraint, implemented via a discriminator, encourages the SRR network to learn anatomically realistic slice structures, improving anatomical fidelity in the final reconstruction. Once trained, SAFFRON enables fast and accurate reconstruction of 3D fetal brain volumes from three orthogonal 2D stacks, offering a robust and efficient solution for clinical and research applications.

B. Contribution

We introduce SAFFRON, a novel framework for high-fidelity 3D fetal brain MRI reconstruction from motion-corrupted thick-slice acquisitions. Our method achieves both computational efficiency and reconstruction quality superior to existing approaches, while specifically addressing the challenges of: (1) large inter-slice spacing (>3 mm) and (2) significant fetal motion artifacts. Our key contributions are:

- 1) We propose a novel reconstruction framework that combines physics-informed, label-free self-supervised learning, guided by a slice acquisition model, with data-driven learning for efficient and accurate reconstruction. The task is reformulated into two decoupled components: a multi-stack motion estimation module, implemented via an SVR network, and a 3D volume reconstruction module, implemented via an SRR network. This design eliminates the need for 3D ground truth data and enables a substantial increase in reconstruction speed by leveraging both model-driven and data-driven advantages.
- 2) To improve the accuracy of motion estimation, we introduce a stack-level contextual consistency loss, which leverages cross-view information to guide the prediction of slice transformations.
- 3) To enhance the anatomical realism of the reconstructed volumes, we incorporate a slice-level adversarial slice structure constraint, encouraging the SRR network to learn and preserve realistic anatomical structures.

We evaluate SAFFRON extensively on both simulated and real clinical datasets, demonstrating that it significantly

outperforms state-of-the-art methods. Notably, our framework achieves a remarkable sixtyfold improvement in reconstruction speed.

II. RELATED WORK

A. Optimization-Based 3D Fetal Brain Reconstruction

Traditional optimization-based methods have long been used for fetal brain MRI reconstruction [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. These techniques, pioneered by Rousseau et al. [5], employ iterative SVR and SRR to generate 3D isotropic volumes from multi-view 2D slice stacks. To address interpolation blurring, [7] and [10] introduced improvements to the point spread function (PSF) interpolation kernel. [7] proposed a Gaussian-weighted PSF kernel to handle slices up to 3 mm thick, although challenges arise with thicker slices (>3 mm). [8] and [10] identified misalignment as a major cause of reconstruction errors and proposed strategies for rejecting misaligned slices. [13] introduced an automated slice-level outlier rejection method, though its performance is heavily reliant on the quality of the initial reconstruction. Despite these advancements, optimization-based methods remain computationally intensive and sensitive to initial conditions, such as poor initialization of the 3D volume or large slice thickness. These issues can lead to suboptimal results, especially in large datasets or when motion corruption is significant.

B. Data-Driven Learning for MRI Reconstruction

Recent progress in data-driven machine learning has improved MRI reconstruction by leveraging latent prior knowledge learned from data. Deep neural networks are increasingly used to extract latent space representations, transforming under-sampled or noisy data into high-quality images [25], [26], [27], [28]. Ref. [25] introduced a CNN that learns the mapping between under-sampled and fully sampled MRI images, significantly improving speed and image quality. Subsequent work using architectures like UNet, ResNet, and recurrent neural networks has further demonstrated success in reducing artifacts, enhancing resolution, and accelerating imaging [27], [29]. In fetal brain volume reconstruction, data-driven approaches typically focus on predicting slice motion parameters [15], [16], [18], [19], [20], [21]. These networks predict the motion of each slice, which is then used for slice alignment and volume reconstruction. For example, [15] employed CNNs to predict motion parameters, while [19] proposed a recurrent network to predict slice motion conditionally based on surrounding slices. [21] used a transformer-based network with attention mechanisms for motion parameter prediction. While promising, these methods are limited by the lack of real 3D fetal brain volumes, leading to training on synthetic datasets. This significantly affects their performance on clinical data.

C. Self-Supervised Learning for MRI Reconstruction

Recent self-supervised methods have made strides in MRI reconstruction by combining forward models with neural networks to reconstruct high-quality images from corrupted, under-sampled, or noisy data, without requiring ground truth images [23], [24], [30]. For instance, [30] used multi-layer

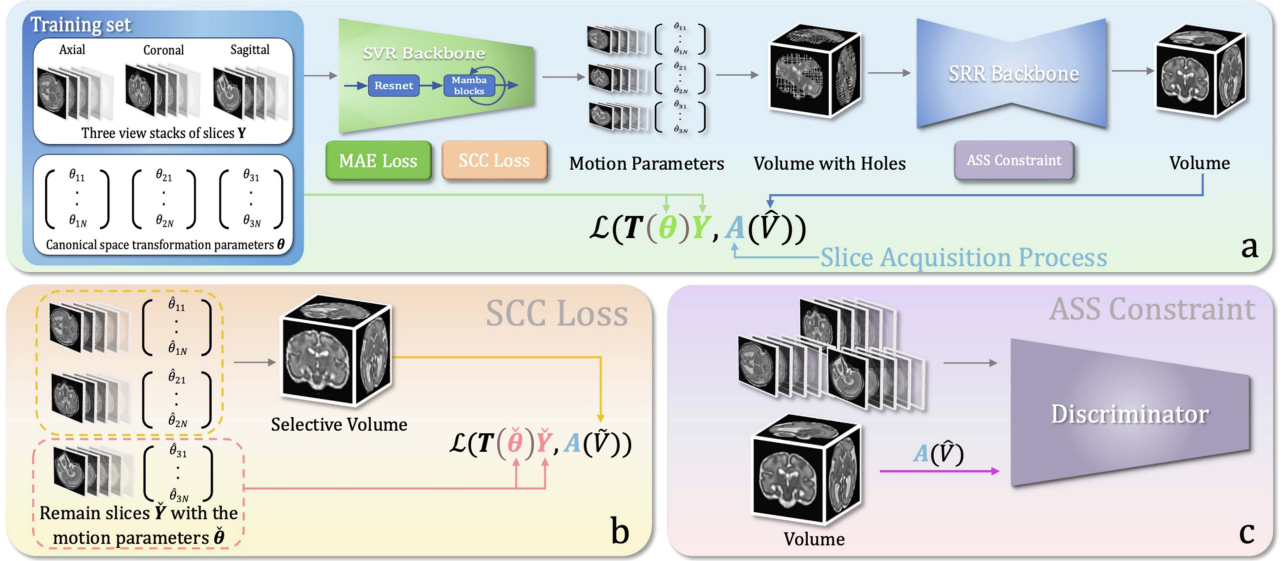


Fig. 1. Overview of our fetal brain MRI reconstruction framework. (a) The framework consists of an SVR backbone for motion estimation and an SRR backbone for 3D volume reconstruction. The SVR network predicts motion parameters for each 2D slice, refined using a stack-level contextual consistency (SCC) loss. The SRR network then reconstructs a high-resolution 3D volume, constrained by slice acquisition-guide SRR loss and adversarial slice structure (ASS) constraint. (b) SCC loss enforces alignment across slice stacks by minimizing discrepancies between different views. (c) ASS constraint enhances anatomical accuracy, ensuring realistic reconstructions.

perceptrons (MLPs) with a physical MRI forward model to reconstruct high-quality MR images from under-sampled k-space data. In fetal brain MRI reconstruction, methods like NeSVoR and ASSURED [23], [24] integrate physical slice acquisition models with MLPs and CNNs to improve reconstruction performance. While these methods show promise, they have limitations. Notably, they do not learn the latent volume space of the reconstruction, which limits their ability to efficiently handle large datasets. Moreover, their motion correction capabilities are limited, making them less suitable for clinical applications where motion artifacts are significant. Motivated by these observations, we clarify our interpretation of “self-supervised” in this work: we adopt the model-based perspective, where the supervisory signal originates from the input data via physics-based acquisition models and intrinsic consistency constraints, rather than from external annotations. This aligns with prior studies [31], [32], [33], [34] where similar training signals are widely described as self-supervised.

III. METHOD

The overall framework of SAFFRON is illustrated in Fig. 1. Our method eliminates the need for 3D ground truth while significantly accelerating reconstruction, combining the strengths of both model-driven and data-driven paradigms. The reconstruction process is divided into two stages: multi-stack motion estimation, handled by an SVR network, and 3D volume reconstruction, performed by an SRR network. To enhance motion estimation, we incorporate a stack-level contextual consistency (SCC) loss, which promotes alignment across multiple stacks. For improved anatomical accuracy during reconstruction, a slice-level adversarial slice structure (ASS) constraint is introduced to guide the SRR network in preserving realistic anatomical features. The **SCC module** is introduced in Section III-C,

and the **ASS constraint** in Section III-D, with their impact quantitatively evaluated in Section V-B (Ablation Study).

1) *Problem Formulation:* Given three 2D MRI slices of stacks, denoted as $Y = \{y_{ij}\}_{i=1,2,3}^{j=1,2,\dots,N}$, our goal is to learn a mapping function $f(Y) = \hat{\theta} = \{\hat{\theta}_{ij}\}_{i=1,2,3}^{j=1,2,\dots,N}$ that estimates the rigid motion parameters for each slice y_{ij} . Here, i indexes the 2D image stacks, while j indexes individual slices within each stack. These motion parameters are represented as a transformation matrix $T(\hat{\theta}_{ij})$, which is given by:

$$T(\hat{\theta}_{ij}) = \begin{bmatrix} R_{ij} & t_{ij} \\ 0 & 1 \end{bmatrix} \quad (1)$$

Here, $\hat{\theta}_{ij} = (r_x, r_y, r_z, t_x, t_y, t_z)_{ij}$, where $T(\hat{\theta}_{ij}) \in \text{SE}(3)$, $R_{ij} \in \text{SO}(3)$, and $t_{ij} \in \mathbb{R}^3$. The translation vector $t_{ij} = (t_x, t_y, t_z)_{ij}$ is defined in Euclidean space, while the rotation matrix R_{ij} , which lies in $\text{SO}(3)$, is parameterized by Euler angles $r_{ij} = (r_x, r_y, r_z)_{ij}$ along the three axes. Using the predicted motion parameters, we stack all 2D MRI slices into an intermediate volume, $\bar{V}$. A 3D volume stacked from three orthogonal slices of stacks may contain gaps or missing regions if the thickness is large. To address this, we input $\bar{V}$ into a CNN-based interpolation network to generate a complete 3D volume $\hat{V}$:

$$\hat{V} = f_{\text{SRR}}(\bar{V}) \quad (2)$$

B. Self-Supervised Data-Driven Learning Strategy

1) *Backbone:* As illustrated in Fig. 1(a), our framework integrates a cascade-based slice-to-volume registration (SVR) backbone, combining a deep residual network (ResNet) [35] with a selective state-space sequence model (Mamba) [36].

This architecture is designed to effectively capture spatiotemporal representations from 2D slice sequences. ResNet first extracts high-level features from individual slices, which are then concatenated and processed by Mamba blocks, enabling sequence modeling for motion transformation prediction. Once these transformations are estimated, the 2D slices are aligned and stacked, forming an intermediate volume that is subsequently refined by the super-resolution reconstruction (SRR) backbone. The SRR network is implemented using a 3D Residual U-Net [37] to generate the final high-resolution 3D volume.

2) *Canonical Space Training Loss*: To reconstruct the 3D volume in a canonical space, we first obtain transformation parameters $T(\theta_{ij})$ using the ITK tool [38]. These parameters serve as supervision for training both the SVR and SRR networks. For the SVR backbone, we minimize the MAE between the predicted motion transformations $T(\hat{\theta}_{ij})$ and the ground truth (GT) canonical transformations $T(\theta_{ij})$ for each slice. The canonical space regularization loss is defined as:

$$L_{SVR} = \sum_{i=1, j=1}^{3, N} \|\hat{\theta}_{ij} - \theta_{ij}\|_1 \quad (3)$$

For the SRR backbone, we apply MAE to measure the difference between the predicted and input slices. The predicted slices are obtained by applying a slice acquisition process $A_{ij}(\cdot)$ to the reconstructed 3D volume. The slice acquisition-guided SRR loss is defined as:

$$L_{SRR} = \sum_{i=1, j=1}^{3, N} \|T(\theta_{ij})\mathbf{y}_{ij} - A_{ij}(\hat{\mathbf{V}})\|_1 \quad (4)$$

3) *Slice Acquisition Model*: The MRI slicing process for one slice $\mathbf{y}_{ij}$, $A_{ij}(\hat{\mathbf{V}})$, can be simulated using a PSF blurring operation [39]. Specifically, the PSF blurring operation on the volume $\hat{\mathbf{V}}$ is approximated by 3D Gaussian filtering. For a given pixel p in the aligned slice $T(\theta_{ij})\mathbf{y}_{ij}$, its intensity $A_{ij}(\hat{\mathbf{V}})(p)$, after PSF blurring, becomes the Gaussian-weighted average of the intensities of its neighboring voxels, p'_s .

$$A_{ij}(\hat{\mathbf{V}})(p) = \frac{1}{S} \sum_{s=1}^S g(\mathbf{u}_s, \Sigma_{ij}) \hat{\mathbf{V}}(p'_s) \quad (5)$$

Here, S is the number of neighboring voxels, and p'_s is derived by $p' = \mathbf{u}_s + p$, with $\mathbf{u}_s$ sampled from a 3D Gaussian distribution, $\mathbf{u}_s \sim \mathcal{N}(\mathbf{0}, \Sigma)$. The Gaussian function $g(\cdot)$ provides the weights, with $\mathbf{u}_s$ as the mean and Σ_{ij} as the variance-covariance matrix:

$$g(\mathbf{u}_s, \Sigma_{ij}) = \frac{1}{\sqrt{(2\pi)^3 \det(\Sigma_{ij})}} \exp\left(-\frac{1}{2} \mathbf{u}_s^T \Sigma_{ij}^{-1} \mathbf{u}_s\right) \quad (6)$$

For SSFSE sequences, the variance-covariance matrix Σ_{ij} is defined as $\text{diag}(\frac{(1.2s_1)^2}{8 \ln(2)}, \frac{(1.2s_2)^2}{8 \ln(2)}, \frac{s_3^2}{8 \ln(2)})$, where s_1 and s_2 represent the in-plane resolution, and s_3 denotes the through-plane resolution [10].

C. Stack Contextual Consistency

Accurately estimating slice transformations within MRI stacks is challenging due to variations in contrast and intensity

caused by fetal movement and different scan orientations. To address this, we introduce an SCC loss, illustrated in Fig. 1, which leverages contextual information from the MRI scan sequence to refine motion estimation. Our approach consists of three main steps. First, the SVR backbone predicts motion parameters $\hat{\theta}$ for all slices across three stacks. Next, two stacks are randomly selected to reconstruct an intermediate volume $\tilde{\mathbf{V}}$. Using the estimated motion parameters $\hat{\theta}$ of the remaining stack, we extract slices $\tilde{\mathbf{Y}} = A_{ij}(\tilde{\mathbf{V}})_{i=1}^{j=1,2,\dots,N}$ that correspond to the unselected stack $\tilde{\mathbf{Y}}$. For accurate motion estimation, the estimated slices $\tilde{\mathbf{Y}}$ should align closely with $\tilde{\mathbf{Y}}$, ensuring consistency across different stacks. To enforce this alignment, we define the SCC loss as the MAE between $\tilde{\mathbf{Y}}$ and $\tilde{\mathbf{Y}}$:

$$\mathcal{L}_{SCC} = \|\tilde{\mathbf{Y}} - \tilde{\mathbf{Y}}\|_1 \quad (7)$$

D. Adversarial Slice Structure Constraint

To promote anatomically realistic reconstruction, we introduce an ASS constraint, which distinguishes reconstructed slices from real ones during training. Since slices from similar brain regions share consistent structural patterns, this constraint enables the model to detect and correct anatomically implausible regions in the reconstructed volume. During each training iteration, the ASS uses a discriminator, D , to compare reconstructed slices $\hat{\mathbf{y}} = \{A_{ij}(\hat{\mathbf{V}})\}_{i=1,2,3}^{j=1,2,\dots,N}$ with real input slices $\mathbf{y} = \{\mathbf{y}_{ij}\}_{i=1,2,3}^{j=1,2,\dots,N}$. This encourages the reconstructed slices $\hat{\mathbf{y}}$ to increasingly resemble the structure of real slices $\mathbf{y}$, as shown in Fig. 1(c). The adversarial loss is divided into two parts:

$$L_d = \mathbb{E}_{\hat{\mathbf{y}} \sim \mathbb{P}_{\hat{\mathbf{y}}}, \mathbf{y} \sim \mathbb{P}_{\mathbf{y}}} \left[D(\hat{\mathbf{y}}) - D(T(\hat{\theta})\mathbf{y}) + \frac{\|D(\hat{\mathbf{y}}) - D(T(\hat{\theta})\mathbf{y})\|_2^2}{2\|\hat{\mathbf{y}} - T(\hat{\theta})\mathbf{y}\|_1} \right] \\ L_g = -\mathbb{E}_{\hat{\mathbf{y}} \sim \mathbb{P}_{\hat{\mathbf{y}}}} [D(\hat{\mathbf{y}})] \quad (8)$$

Here, the third term in L_d is the quadratic potential divergence from [40], which stabilizes the model without requiring additional Lipschitz constraints.

IV. EXPERIMENTS SETUP

We conducted comprehensive evaluations of our proposed SAFFRON framework using both simulated and clinical fetal brain MRI datasets to assess its performance under realistic motion conditions.

A. Simulation Dataset

We evaluated SAFFRON on two publicly available fetal brain MRI datasets, which serve as ground truth (GT) references for comparison: 1) **FeTA dataset** [41]: comprises 80 fetal brain MRI volumes, with 68 used for training and 12 for testing. 2) **FBA dataset** [42]: consists of 14 high-resolution (0.8 mm isotropic) fetal brain MRI volumes, split into 7 for training and 7 for testing. For training, we combined 68 FeTA and 7 FBA volumes, resulting in a 75-volume dataset. Model performance was then evaluated on two separate test sets (12 FeTA and 7 FBA volumes). In addition to the FeTA and FBA datasets, we constructed a Clinical Simulated (CS) dataset by reconstructing high-quality fetal brain volumes from minimal-motion clinical scans using the NiftyMIC pipeline. These volumes served as

pseudo-ground truth for generating simulated multi-stack input slices. The CS dataset comprises 160 volumes (100 for training and 60 for testing).

1) *Simulation Strategy*: To generate training data, we simulated three-view orthogonal stacks with realistic motion corruption. We simulate three orthogonal stacks (axial, sagittal, coronal) to reflect typical clinical fetal brain MRI protocols [3], [43], [44], [45], where scan time and motion constraints limit the number of acquisitions. Although SAFFRON supports more stacks, this design choice ensures that our evaluation aligns with realistic clinical practice. GT volumes, aligned to a fetal brain atlas [45] and resampled to a 0.8 mm isotropic resolution, served as the reference. To introduce motion variability, we applied random horizontal flips, rotations, and translations. Each subject's dataset included three orthogonal stacks with a 0.8 mm in-plane resolution and a slice thickness of 4-5 mm. To simulate motion corruption, we introduced random rotations (Euler angles: $\pm 90^\circ$) and translations (± 6 mm) using a slice acquisition process. Additionally, we mimicked twoshot sequences, commonly used in 2D MRI, by interleaving motion-corrupted slice halves. Fetal brain motion trajectories were modeled as random walks with motion offsets [46], [19], [20]. Using these trajectories, we adjusted the volumes and extracted corresponding slices. In addition to rigid displacements, we simulated fetal MRI-specific motion artifacts, including motion-induced blurring, signal dropout, bias field inhomogeneity, and contrast jitter, following the approaches described in [47], [48]. These augmentations improve robustness against typical fetal MRI distortions.

For the FeTA test dataset, motion parameters were sampled with 3D rotations uniformly from $U(-3, 3)$ degrees and 3D translations from $U(-2, 2)$ mm. For the FBA test dataset, motion parameters were sampled from $U(-18, 18)$ degrees for rotation and from $U(-4, 4)$ mm for translation. For the CS test dataset, 3D rotations were sampled from $U(-16^\circ, 16^\circ)$ and 3D translations from $U(-4, 4)$ mm.

B. Clinical Dataset

The clinical dataset used in this study was retrospectively collected from archived MRI scans acquired on a 1.5 T Siemens Magnetom Aera scanner (Syngo MR E11) using an eight-channel body coil and a T2-weighted Half-Fourier Acquisition Single-shot Turbo spin Echo (HASTE) sequence. The scanning parameters were TR/TE = 1400/92 ms, in-plane resolution = 1.172×1.172 mm, and slice thickness = 4-5 mm. The dataset included MRI scans from 400 fetuses, with gestational ages (GA) ranging from 22 to 35 weeks. For training, we used 300 subjects (900 stacks), while the test set comprised 100 subjects (300 stacks). This dataset consisted of 400 retrospectively collected fetal brain MRI scans acquired during routine clinical examinations. Subjects were primarily anatomically normal fetuses referred for MRI following inconclusive ultrasound screening. Cases with severe artifacts or confirmed major neuropathologies were excluded. The selected cohort covers a broad gestational age range (22-36 weeks) and diverse acquisition conditions. This study was approved by the Ethics Committee of the International Peace Maternity and Child Health Hospital under protocol number 2023018.

1) *Data Preprocessing for Clinical Fetus MRI Data*: Preprocessing followed a standardized pipeline. First, a trained segmentation network [13] extracted the fetal brain from the uterine MRI scans. Next, bias field correction [38] was applied to reduce intensity inhomogeneities in each slice. Finally, the ITK tool [38] was used to register each slice to the atlas space, generating canonical transformation parameters for further processing.

C. Baseline and Evaluation Metric

We compared SAFFRON with four state-of-the-art (SOTA) fetal brain reconstruction methods, each with open-source implementations: 1) AutoSVRTK, based on the MONAI-compatible post-processing pipeline available at <https://github.com/SVRTK/auto-proc-svrtk>, which builds upon the SVRTK framework. We follow the recommended citations for this pipeline [10], [14], 2) NiftyMIC, a classical reconstruction framework [13], 3) NeSVoR, an reconstruction method based on implicit neural representations [23], and 4) SVoRT+NeSVoR, an approach utilizing SVoRT [21] to provide initialized transformations for NeSVoR [23]. To ensure a fair comparison, we performed hyperparameter tuning for each method. A randomly selected subject from the simulated fetal dataset was used for tuning, optimizing the hyperparameters to minimize mean squared error (MSE) between the reconstructed volume and the GT. The optimized hyperparameters were then fixed and applied to all test datasets. For evaluation, we assessed the accuracy of volume reconstruction by comparing the reconstructed volumes with the GT using several quantitative metrics, including Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Normalized Root Mean Squared Error (RMSE).

D. Implementation Detail

We implemented SAFFRON using PyTorch and conducted training on an NVIDIA RTX Titan GPU. The model was optimized using the Adam optimizer with a cosine annealing learning rate scheduler. The learning rate was initialized at 0.0001 and gradually reduced, with training conducted for a total of 2×10^5 iterations. An outlier rejection module is used during inference to identify and exclude slices with low similarity to the projected reconstructions, reducing the impact of extreme motion artifacts.

V. EXPERIMENTS AND RESULTS

A. Simulation Experiments

Table I compares the performance of SAFFRON with four baseline methods. AutoSVRTK and NiftyMIC are iterative SVR-SRR methods that rely on good initial spatial alignment between slices and the initial SRR volume. However, their performance declines markedly when the initial SRR volume is reconstructed from slices with substantial motion corruption (defined as inter-slice translations exceeding 4 mm or rotations greater than 12°), which disrupts the spatial consistency required for effective SVR alignment. This effect can be observed in

TABLE I

QUANTITATIVE COMPARISON OF FETAL BRAIN RECONSTRUCTION ACROSS THREE DATASETS. METRICS INCLUDE PSNR, SSIM, AND NRMSE. THE BEST PERFORMANCE IN EACH CATEGORY IS HIGHLIGHTED IN **BOLD**. FeTA AND FBA DATASETS USE KNOWN HIGH-RESOLUTION VOLUMES AS GROUND TRUTH; THE CS DATASET IS DERIVED FROM MINIMAL-MOTION CLINICAL VOLUMES.

Method	FeTA			CS (Clinical Simulated)			FBA		
	PSNR $\uparrow$	SSIM $\uparrow$	NRMSE $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	NRMSE $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	NRMSE $\downarrow$
NiftyMIC	21.24 $\pm$ 1.61	0.77 $\pm$ 0.03	0.08 $\pm$ 0.02	11.55 $\pm$ 2.88	0.35 $\pm$ 0.13	0.30 $\pm$ 0.03	11.07 $\pm$ 3.05	0.34 $\pm$ 0.32	0.29 $\pm$ 0.10
AutoSVRTK	21.25 $\pm$ 1.29	0.79 $\pm$ 0.04	0.07 $\pm$ 0.01	15.23 $\pm$ 2.07	0.41 $\pm$ 0.07	0.17 $\pm$ 0.02	12.46 $\pm$ 3.16	0.36 $\pm$ 0.28	0.25 $\pm$ 0.08
NeSVoR	22.72 $\pm$ 1.30	0.75 $\pm$ 0.03	0.06$\pm$0.01	11.96 $\pm$ 2.09	0.36 $\pm$ 0.11	0.25 $\pm$ 0.02	11.86 $\pm$ 3.09	0.35 $\pm$ 0.29	0.27 $\pm$ 0.09
SVoRT+NeSVoR	22.71 $\pm$ 1.20	0.75 $\pm$ 0.03	0.06$\pm$0.01	22.01 $\pm$ 1.87	0.78 $\pm$ 0.04	0.06$\pm$0.01	19.41 $\pm$ 3.40	0.71 $\pm$ 0.13	0.11 $\pm$ 0.04
SAFFRON	23.19$\pm$1.49	0.81$\pm$0.03	0.06$\pm$0.01	22.52$\pm$1.61	0.81$\pm$0.04	0.06$\pm$0.01	21.84$\pm$4.08	0.80$\pm$0.06	0.08$\pm$0.06

the simulation results on the FBA and CS datasets (Table I). While these methods perform well on the FeTA dataset, they struggle with more challenging datasets, particularly those with severe motion artifacts. NeSVoR, which leverages implicit neural representations, demonstrates strong image-fitting capabilities and effectively captures fine image details. However, its performance is highly dependent on the initialization of the transformation matrix. When tested on the FBA dataset, which exhibits greater motion corruption, its accuracy declines sharply. SVoRT+NeSVoR, which integrates SVoRT for transformation matrix initialization, improves reconstruction on datasets with severe motion artifacts. However, it remains highly sensitive to noise, making it less effective on noisy fetal brain MRI scans. In contrast, SAFFRON consistently outperforms all baseline methods across both datasets. This is attributed to its data- and physics-driven learning strategy, which integrates both data priors and physical priors to enhance reconstruction accuracy. Additionally, our method remains stable across varying degrees of motion. For instance, PSNR and SSIM values remain above 20.00 and 0.80, respectively, for rotational motion corruption ranging from $\pm 3^\circ$ to $\pm 18^\circ$ across both test datasets. These results highlight the superior performance of our approach over SOTA techniques on simulated data.

Qualitative results in Fig. 2 further validate these findings. NeSVoR and SVoRT+NeSVoR produce noisier reconstructions due to overfitting, as they capture image noise along with structural features. AutoSVRTK and NiftyMIC perform well on the FeTA dataset but fail to reconstruct volumes affected by significant motion corruption, resulting in severe artifacts. Notably, NiftyMIC employs an outlier rejection strategy that removes slices with low scores, leading to large gaps in the reconstructed volumes. In contrast, SAFFRON delivers consistently high-quality fetal brain MRI reconstructions across both datasets, reinforcing its robustness and effectiveness. Finally, Fig. 3 compares the reconstruction speeds of all methods on a single NVIDIA RTX Titan GPU. Our method is over $60\times$ faster than NeSVoR, demonstrating a significant improvement in computational efficiency.

B. Ablation Study

1) *Effect of Stack Contextual Consistency*: To evaluate the impact of the SCC module, we conducted an experiment where SCC was removed while keeping all other settings unchanged for a fair comparison. Fig. 4 illustrates the estimated motion

TABLE II

ABLATION EXPERIMENTS OF SCC MODULE ON THE FBA DATASET

Methods	MT $\downarrow$ (mm)	MR $\downarrow$ (degree)
w/ SCC	0.59$\pm$1.84	2.38$\pm$0.96
w/o SCC	1.11 $\pm$ 2.21	4.87 $\pm$ 1.95

TABLE III

ABLATION STUDY OF THE SRR MODULE. QUANTITATIVE COMPARISON OF DIFFERENT SRR CONFIGURATIONS USING PSNR, SSIM, AND NRMSE METRICS. “w/ ASS” REFERS TO THE FULL MODEL WITH THE ADVERSARIAL SLICE STRUCTURE CONSTRAINT; “w/o ASS” EXCLUDES THE ADVERSARIAL CONSTRAINT; AND “w/o CNN” REPLACES THE CNN-BASED INTERPOLATION WITH TRADITIONAL PSF-BASED INTERPOLATION. RESULTS DEMONSTRATE THE CONTRIBUTION OF BOTH CNN ARCHITECTURE AND ADVERSARIAL LOSS TO THE RECONSTRUCTION QUALITY.

Method	PSNR $\uparrow$	SSIM $\uparrow$	NRMSE $\downarrow$
w/ ASS	21.84$\pm$4.08	0.80$\pm$0.06	0.08$\pm$0.06
w/o ASS	19.45 $\pm$ 4.35	0.75 $\pm$ 0.13	0.12 $\pm$ 0.05
w/o CNN	18.79 $\pm$ 3.57	0.70 $\pm$ 0.07	0.13 $\pm$ 0.03

parameters. Without SCC, the method fails to accurately estimate motion parameters, leading to significant offsets. In contrast, with SCC included, the estimated motion parameters closely align with the GT. However, our model exhibits limitations in predicting motion for slices with minimal brain structures, such as the initial and final slices in a stack, which may affect overall performance. Table II presents the quantitative results, where MAE is used to evaluate rotational (MR) and translational (MT) motion estimates. The results show that incorporating SCC significantly improves accuracy, reducing rotation errors by 2.49 degrees and translation errors by 0.52 mm. This study highlights the critical role of SCC in enhancing fetal MRI motion estimation.

2) *Effect of Adversarial Slice Structure Constraint*: We further assess the contribution of the ASS constraint to the super-resolution interpolation process. To isolate its effect, we performed ablations removing the ASS module and replacing the CNN-based interpolation with a traditional PSF-based interpolation, similar to classical SVR methods [9], [10], [12], [13], [23], [24]. Fig. 5 shows that removing ASS leads to blurry anatomical features, while omitting the CNN architecture results in noisy outputs. In contrast, incorporating both CNN-based interpolation and ASS yields sharper and more anatomically faithful reconstructions. Quantitative results in Table III confirm these findings. This study validates that the combination of CNN

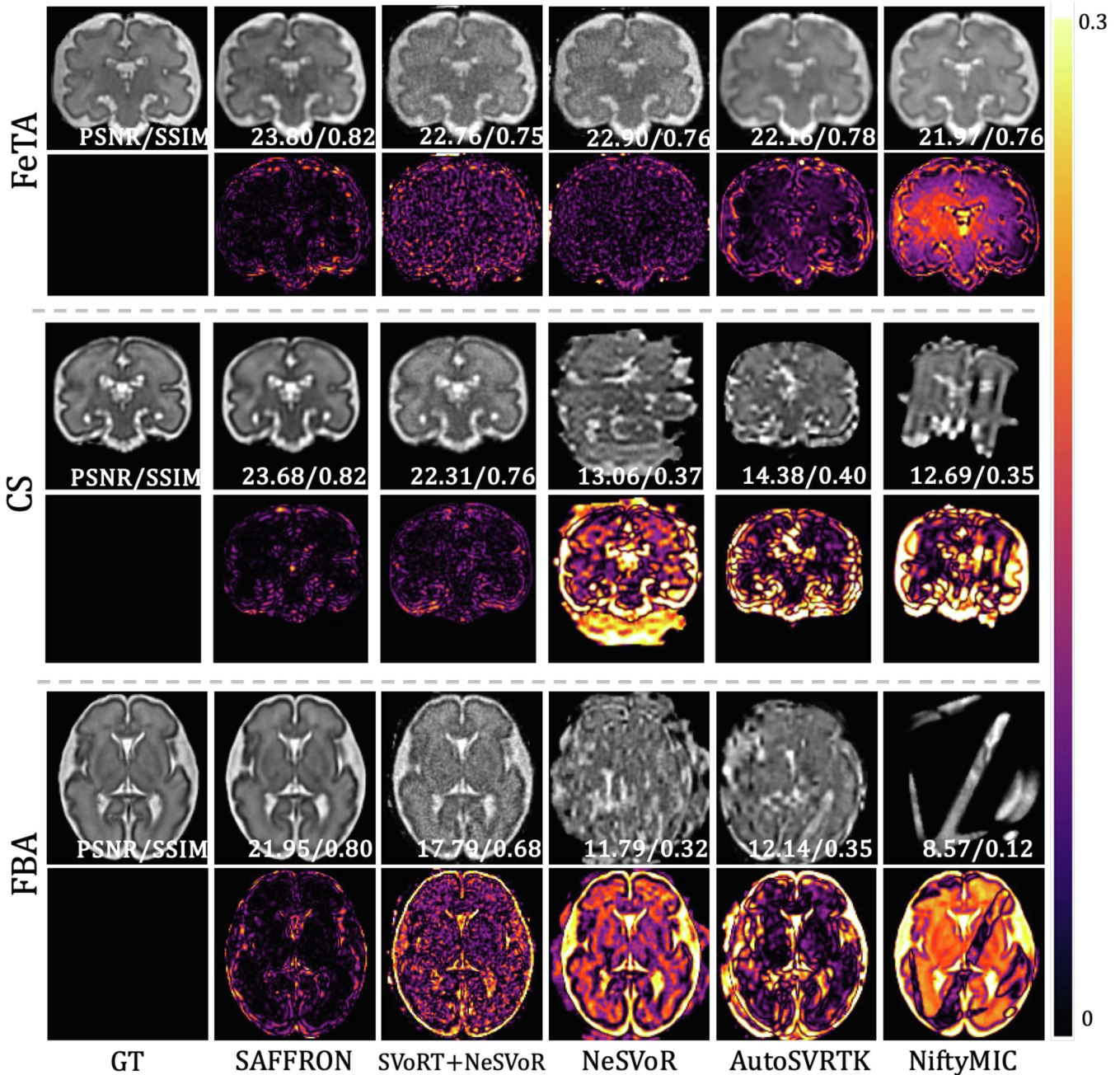


Fig. 2. Qualitative comparison of reconstructed fetal brain volumes on three datasets. Top-to-bottom: FeTA (1st–2nd row), CS (3rd–4th row), and FBA (5th–6th row) datasets. Odd rows show representative reconstructed slices for a fetal subject using different methods. Even rows show the corresponding absolute error maps, computed as the voxel-wise absolute difference between the reconstruction and ground truth (or pseudo-ground truth for CS).

interpolation and structural adversarial constraints significantly enhances the quality and realism of the reconstructed volumes.

3) *Impact of Network Architecture:* Our SAFFRON framework consists of both an SVR backbone and an SRR backbone. To examine the effect of different network architectures on reconstruction performance, we replaced the SVR backbone with a Transformer [49], keeping all other parameters identical. Table IV presents the quantitative results, showing that Mamba achieves comparable performance on both metrics. Similarly, we replaced the SRR backbone with a 3D U-Net [50], and the results in Table V demonstrate that the Residual U-Net-based

TABLE IV
ABLATION EXPERIMENTS OF SVR BACKBONE ON THE FBA DATASET

SVR Backbone	MT↓(mm)	MR↓(degree)
Mamba	0.59±1.84	2.38±0.96
Transformer	0.60±1.89	2.40±0.98

model achieves superior performance. Overall, our findings indicate that network architecture has a negligible impact on fetal brain reconstruction quality. This confirms that the performance improvements in SAFFRON are primarily driven by its

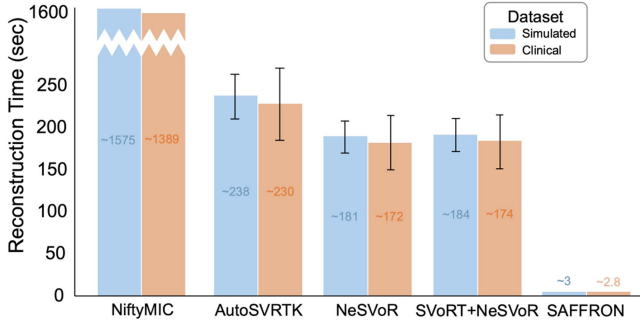


Fig. 3. Runtime comparison on both simulated and clinical datasets. The y-axis shows average reconstruction time per subject (in seconds). Error bars represent standard deviation computed from 100 samples per method. Clinical reconstructions averaged 2.8 seconds (std: 0.9 s).

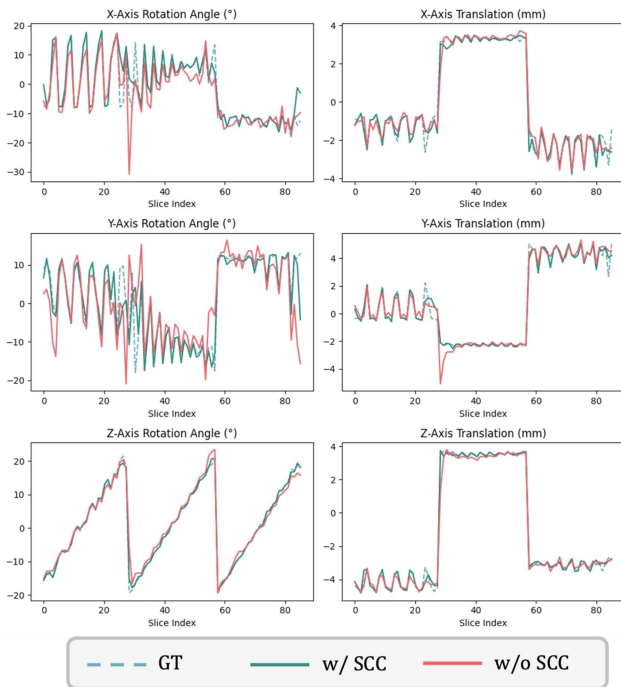


Fig. 4. Qualitative ablation study of motion estimation on the FBA dataset [42]. The full SAFFRON model outperforms models without SCC in predicting motion parameters.

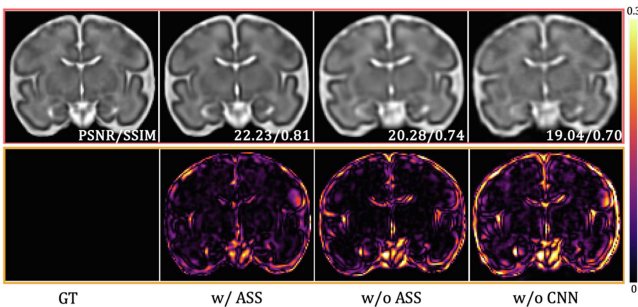


Fig. 5. Qualitative ablation study using a sample from the FBA dataset. The full SAFFRON model delivers superior results compared to models without adversarial slice structure constraint (w/o ASS) and without CNN-based interpolation.

TABLE V
ABLATION EXPERIMENTS OF SRR BACKBONE ON THE FBA DATASET

SRR Backbone	PSNR $\uparrow$	SSIM $\uparrow$	NRMSE $\downarrow$
Residual U-net	21.84$\pm$4.08	0.80$\pm$0.06	0.08$\pm$0.06
U-net	21.66 $\pm$ 4.10	0.80 $\pm$ 0.06	0.08 $\pm$ 0.06

TABLE VI
COMPARISON OF MOTION ESTIMATION ACCURACY ON THE FBA DATASET

Method	MT (mm) $\downarrow$	MR (deg) $\downarrow$
SAFFRON (w/ SCC)	0.59$\pm$1.84	2.38$\pm$0.96
SAFFRON (w/o SCC)	1.11 $\pm$ 2.21	4.87 $\pm$ 1.95
ITK	0.72 $\pm$ 2.05	3.21 $\pm$ 1.35

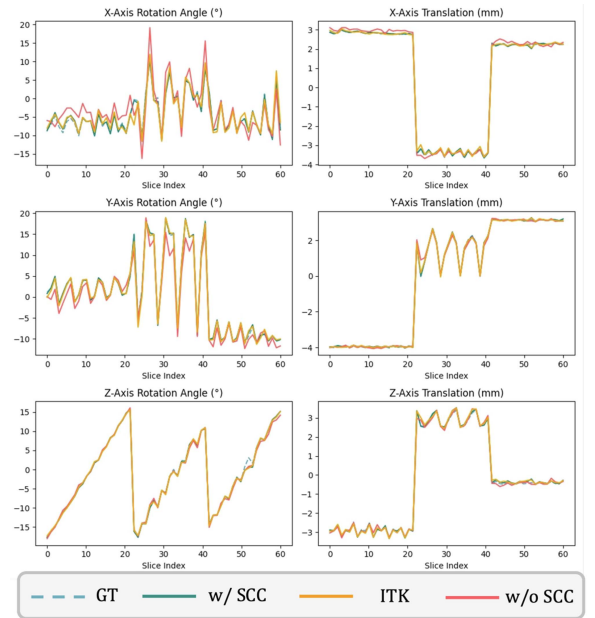


Fig. 6. Qualitative comparison of motion fields estimated by ITK vs. SAFFRON on the FBA dataset.

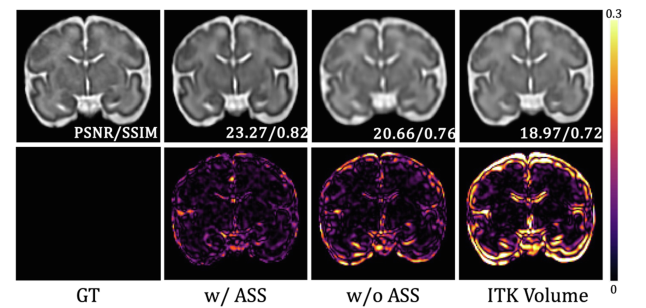


Fig. 7. Qualitative comparison on the FeTA dataset: SRR reconstructions with and without ASS, and using ITK-aligned volumes. Bottom row shows corresponding error maps.

data-driven and self-supervised learning strategy rather than specific architectural choices. We added a component-level evaluation comparing ITK initialization, ITK+SRR, and the full SAFFRON pipeline. As shown in Table VI and Fig. 6,

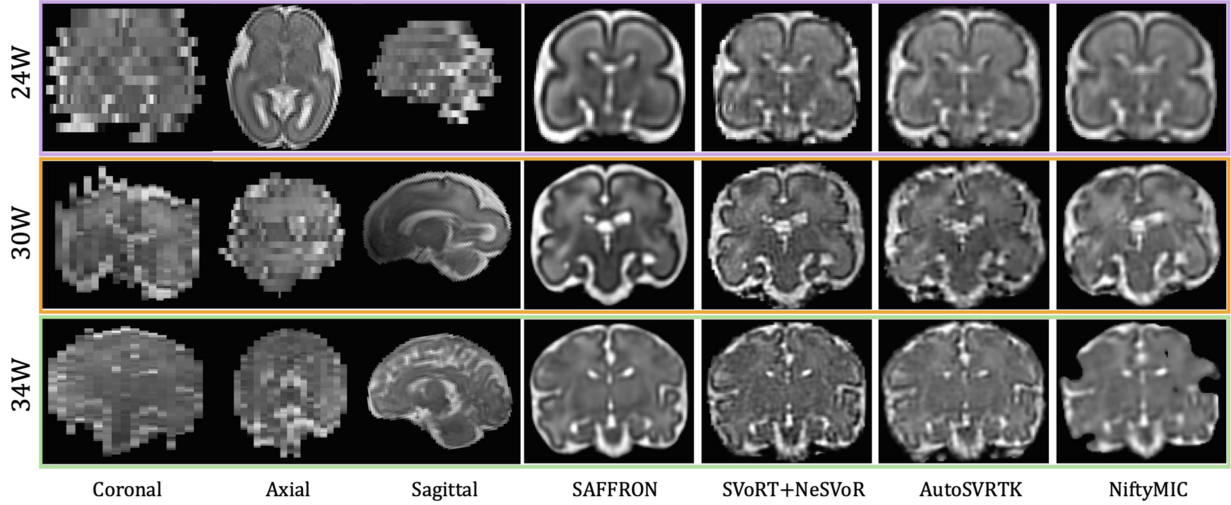


Fig. 8. Reconstructions on clinical data. Displayed are the brain reconstructions of three fetal subjects with gestational ages ranging from 24 to 34 weeks. For each subject, input images (top three columns) are shown in three orthogonal planes: axial, sagittal, and coronal.

TABLE VII
SRR MODULE EVALUATION USING ITK-DERIVED MOTION PARAMETERS
(FETA SIMULATION, 3 STACKS)

Method	PSNR $\uparrow$	SSIM $\uparrow$	NRMSE $\downarrow$
SRR (w/ ASS)	21.84$\pm$4.08	0.80$\pm$0.06	0.08$\pm$0.06
SRR (w/o ASS)	19.45 $\pm$ 4.35	0.75 $\pm$ 0.13	0.12 $\pm$ 0.05
ITK Volume	18.52 $\pm$ 4.94	0.71 $\pm$ 0.16	0.13 $\pm$ 0.06

SAFFRON achieves improved motion estimation accuracy beyond the ITK baseline. Furthermore, Table VII and Fig. 7 demonstrate that the SRR module—particularly when combined with ASS—substantially enhances reconstruction quality even when applied to ITK-aligned slices.

C. Clinical Experiments

We evaluated our method on 100 clinical fetal brain MRI subjects from the test set. Fig. 8 provides a qualitative comparison of 3D volume reconstructions produced by our approach and several SOTA models. NeSVoR was excluded from the comparison as it does not reconstruct volumes in the canonical atlas space without SVoRT initialization. Qualitatively, our method reconstructs key anatomical structures—including the cerebrospinal fluid (CSF), gray matter (GM), and white matter (WM)—with better continuity and fewer artifacts. In contrast, SVoRT+NeSVoR introduces more noise and intensity inconsistencies, likely due to the sensitivity of implicit neural representations (INR) to noise. AutoSVRTK and NiftyMIC exhibit more artifacts, as they struggle to fully correct motion-induced distortions. While these methods employ outlier rejection modules, they can sometimes degrade performance by mistakenly discarding useful slices in complex cases. In comparison, our method consistently achieves superior reconstruction quality, preserving finer anatomical details and effectively reducing noise. Since GT reconstructions are not available for clinical fetal brain data, we conducted a qualitative assessment (QA) on 100 fetal brain cases

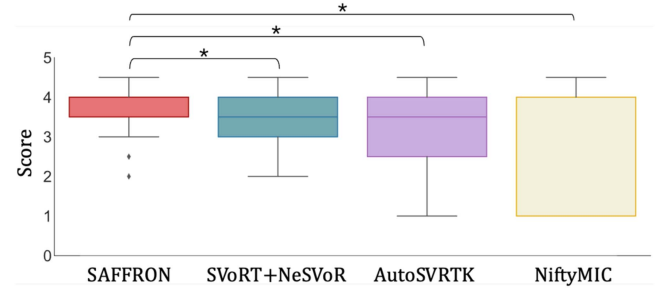


Fig. 9. QA scores of different methods on the clinical fetal dataset. Results of Wilcoxon signed rank test are presented, * : $p < 0.05$ and n.s.: not significant ($p \geq 0.05$).

to evaluate the perceptual quality of the reconstructed volumes. The qualitative assessment was conducted by two board-certified radiologists with over 5 years of experience in fetal MRI. Each rater scored 100 reconstructed fetal brain volumes using a 5-point scale, blinded to reconstruction method. Inter-rater reliability was measured using the weighted Cohen's kappa ($\kappa_w = 0.92$), indicating almost perfect agreement. For cases where scores differed by more than one point, a consensus score was reached via discussion. Volumes receiving a score greater than 3 were considered suitable for further clinical analysis. Fig. 9 demonstrates that SAFFRON consistently outperforms baseline models, achieving higher image quality across all test cases.

D. Clinically-Oriented Application: Growth Charts of Normal Fetal Brain Development

To evaluate the clinical applicability of our fetal brain reconstruction method, we tested its performance on a clinically relevant growth chart analysis that tracks fetal brain development. Specifically, we generated volumetric growth charts for major brain structures, focusing on total deep GM (Fig. 10), covering GAs between 23 and 34 weeks. The study included

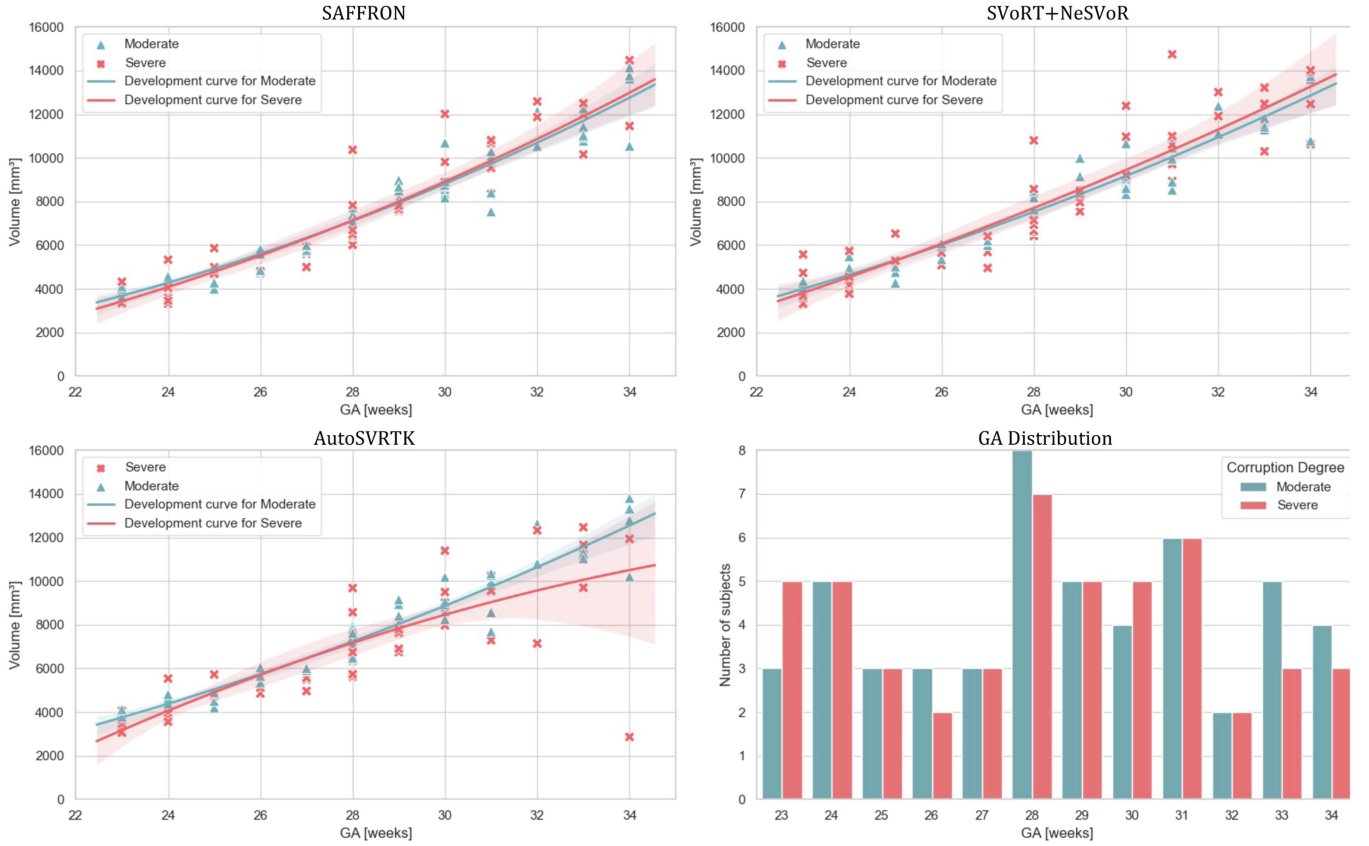


Fig. 10. Growth chart: volumetry of the major brain structure (total deep GM) for two fetal cohorts from clinical fetal MRI with different reconstruction methods.

TABLE VIII
THE SIGNIFICANCE TEST. RESULTS OF THE T-TEST AND MAE OF TWO DEVELOPMENT CURVES ARE PRESENTED.

Methods	P-value	MAE
SAFFRON	0.759	98.21
SVoRT+NeSVoR	0.006	190.74
AutoSVRTK	<0.001	555.31

51 cases with moderate motion corruption and 49 cases with severe motion corruption. Anatomical segmentations for computing growth chart metrics were generated using the pretrained BOUNTI segmentation model [51], which automatically parcellates fetal brain structures such as the cortical plate and ventricles. To assess motion-related effects, we computed the median translational and rotational motion parameters derived from the SVR-estimated transformations for each subject. Subjects were grouped into motion categories based on median translation (up to 4 mm/s) and rotation (up to 6°/s). For each reconstruction method, we generated two developmental curves from the volumetric growth charts and compared their similarity. A high degree of similarity between the curves indicates greater robustness in fetal brain reconstruction. To statistically assess the differences between these curves, we conducted a Student's t-test, with the null hypothesis stating that no significant difference exists between the curves for each method. As shown in Table VIII, the t-test results revealed p-values

< 0.05 for both SVoRT+NeSVoR and AutoSVRTK, indicating significant discrepancies in their reconstructed growth curves. In contrast, our method demonstrated greater robustness, as the development curves remained statistically consistent, even in cases with severe motion corruption. These findings confirm that our approach provides more reliable fetal brain reconstructions, particularly in challenging clinical scenarios where motion artifacts are prevalent. In this context, the use of growth charts is not intended to establish clinical claims regarding neurodevelopment or diagnosis. Instead, the analysis is designed to assess reconstruction robustness by examining structural volume trajectories across varying levels of motion corruption. The observed consistency of these trajectories under high-motion conditions indicates that SAFFRON preserves anatomical plausibility more effectively than baseline methods.

VI. DISCUSSION AND CONCLUSION

In this work, we introduced SAFFRON, a novel framework that combines self-supervised learning, guided by a slice acquisition model, with data-driven strategies for efficient and accurate 3D fetal brain MRI reconstruction. To overcome the limitations of traditional iterative methods, we reformulated the reconstruction task into two modular components: (1) multi-stack motion estimation, using an SVR network to align 2D slices to a canonical space via rigid transformations, and (2)

3D volume reconstruction, using an SRR network to generate high-resolution volumes from the aligned slices. Unlike supervised approaches that require labeled 3D ground truth, SAFFRON is trained exclusively on 2D slices, using a differentiable slice acquisition model. To further enhance performance, we introduce a stack-level contextual consistency loss to improve motion estimation and a slice-level adversarial constraint to enforce anatomical realism. Together, these components guide the SVR and SRR networks toward more accurate and consistent reconstruction. Extensive evaluations on both benchmark and clinical datasets demonstrate that SAFFRON outperforms existing state-of-the-art methods, delivering a $60\times$ improvement in reconstruction speed without sacrificing accuracy.

Despite these strengths, SAFFRON has certain limitations. While SAFFRON shows strong resilience to typical motion artifacts, its performance may degrade under extreme motion scenarios beyond the distribution seen in training. In such cases, the outlier rejection module provides a fail-safe mechanism, though some quality loss is still possible. Currently, it focuses on rigid motion correction, whereas real-world fetal motion often includes complex non-rigid deformations, such as those caused by fetal organ or body movement. Incorporating non-rigid motion modeling—potentially through methods like DSVRTK [52]—is a promising avenue for future work. Additionally, while SAFFRON achieves strong performance with single-modality T2-weighted MRI, extending the framework to multi-contrast reconstruction could further enhance image fidelity and clinical utility. Future work may also explore its application to other anatomical regions and imaging modalities to broaden its scope and impact.

In clinical fetal MRI acquisition, uncertainties such as unpredictable fetal motion, maternal breathing, and scanner-specific distortions can introduce imaging artifacts that deviate from those generated in simulated training environments. This domain gap may introduce bias in reconstruction quality, especially when the model encounters out-of-distribution corruptions. To reduce this bias, we plan to refine our artifact simulation pipeline using real clinical scan statistics, including modeling of noise, bias fields, and motion trajectories. Moreover, we propose to integrate uncertainty-aware learning strategies, where the SVR network is extended to estimate slice-wise uncertainty and penalize high-uncertainty slices during reconstruction. This approach may enhance the model's robustness to clinical artifacts not seen during training.

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